

same and that is why it can drive autonomously in few cities. Their level 3 car can drive autonomously at almost every place in the world where it is deployed right now, whereas other companies are still limited to a few cities. That is because, in order to expand to other cities, they will have to collect more and more data.

Second, since we do not use lidar, we don't need lidar data. And that reduces the data dependency further. This is how a major chunk of the data dependency is removed.

The next thing is that, normally other companies need continuous data when driving from one point to another. But in our case, we do not need continuous data, we need specialised data. For example, we need a couple of minutes in typical Delhi traffic or discontinuous data from different places. In a multi-task approach, other companies collect all the data like sensor data, camera data, and other data for ultrasonic radar, IMUs, and GPS, etc.

This is where our AI, which is less dependent on data, comes in. We only need camera data and that too for particular tasks like object detection, segmentation, etc. Instead of predicting things with a neural network, these are calculated from the output of the first few subtasks. So, we need data for only five tasks instead of all tasks because the tech is able to deduce things like distance of a car and its size using the 2D data with the help of certain proprietary algorithms. So, it uses those kinds of heuristics so that instead of predicting everything, certain things can be calculated, which is computationally efficient.

Our system decreases data usage in three ways. You don't need lidar data, you need selective data due to multi-task learning, and our proprietary AI is less dependent on data, which can extract the maximum of information from a limited amount of data. We are in the process of filing patents for the same.

While for the demo you used a gaming laptop, will you be developing any specialised electronics hardware in the near future?

We are a technology company. We will be sourcing electronics hardware with custom configurations through supply chain partnerships.

Can you shed some light on how you reduce the number of cameras and radars?

We use eight cameras to get a 360-degree view of the surroundings and

prepare a bird's eye view. We have the size and acceleration of all the objects that get detected by the cameras, and get their vertical and horizontal speed from 2D data only. Following this we reconstruct a map from an aerial view, which has different objects plotted on it, and then we plan trajectories to move our vehicle forward. All of these things are done by our software.

Basically, your system works on a reconstruction where you capture the image of the objects from different cameras to get the 3D data. Is that correct?

Yes, there are eight cameras at different locations to generate a 360-degree view. Each camera has a certain view angle, so when you combine all the feed of the cameras together, it gives you a 360-degree video. From a 360-degree horizontal video and that data, we can extract information like speed and distance of the non-ego vehicles. But for each vehicle, this information is captured using a single camera only. This is our first patent, which is getting the dynamics of all the entities around you. From a single camera, you can detect the vertical or horizontal speed, acceleration, size, and its trajectory, which in turn is used to plan our own vehicle's trajectory.

Are you using any sensors and radars?

Yes, we are also using radar and ultrasonics as backup sensors. These are not active sensors but passive sensors to optimize the output from the cameras. For our own localisation, we use GPS and IMU, which are necessities because you need to know the velocity of your own vehicle also and these are comparatively cheaper. The sensors basically removed by us are just lidars to reduce the expense. Overall, we have some radars and sensors as backups to ensure that the tech without lidar is safer than the with-lidar tech.

I must add that while the tech itself is cool, safety is our primary goal. We don't need self-driving cars because they are cool but because there are too many accidents due to human negligence. We need to replace humans with something safer.

What is your proposed business model?

We are starting with driverless fleets for micro-mobility in private institutions like educational campuses, industrial parks, theme parks, airports, etc. Post that, we will be entering into

public-road driverless shared mobility solutions in partnership with OEMs. Singapore will serve as our first target market due to favourable regulatory infrastructure.

Usually, for a ride, the company takes twenty percent of the ride charge as commission while the remaining eighty percent goes to the drivers. But in our case, we have no drivers, and keeping in mind the maintenance and charging costs, we have a better return on investment than normal car-hailing services.

Are we correct to understand that you might be forced to launch a driverless fleet in Singapore rather than India?

We'll be ready for India by 2023, but the lack of regulations is what is stopping us from entering it right away. Now, these regulations are not a one-person show. Any moment you are making a disruptive change, that can't be done by one person or company. It is a collaborative effort of the automotive companies, major OEMs, regulatory bodies, and the media particularly, and which is why I take every opportunity to give that message out.

But things don't end here. We'll be starting with a smaller fleet for private campuses like educational institutions, industrial parks, theme parks, etc, and we'll be starting with our beta testing in these campuses within a year. Post that, we will be entering Singapore for public road testing and deployment.

Why do you want to shift to Bengaluru when you are studying in Punjab?

The community and the people working for us are mostly from that area, and also because Bengaluru provides better industry connections. Since we will be partnering a lot, Punjab is not the ideal place for setting up an industry as technical resources available there are limited.

There are also other factors that add to the costs and eat away most of your capital. The Karnataka government is very supportive of startups. They have been very supportive of electric vehicles, and we really hope that it will be the same for autonomous vehicles as well. Moreover, there are large aerospace parks in Bangalore where even space-tech companies test their engines; we need that kind of space to test our vehicles as well.

Are you inviting investments?

Yeah, we are open to it. **EFY**